Practical-6
Rendering External JSON Data using Fetch API, Search and Filter

Theory
1. How is JSON fetched, parsed, and rendered?
Answer: JSON is fetched using the Fetch API. First, JavaScript uses fetch() to get the JSON file. Then the response is converted into JavaScript data using .json(). After parsing, the data is used to create HTML elements and display the information on the webpage.
So, the basic process is:
Fetch JSON → Parse JSON → Use Data → Render on Webpage

2. Which array methods are used for search, filter, sort, and pagination?

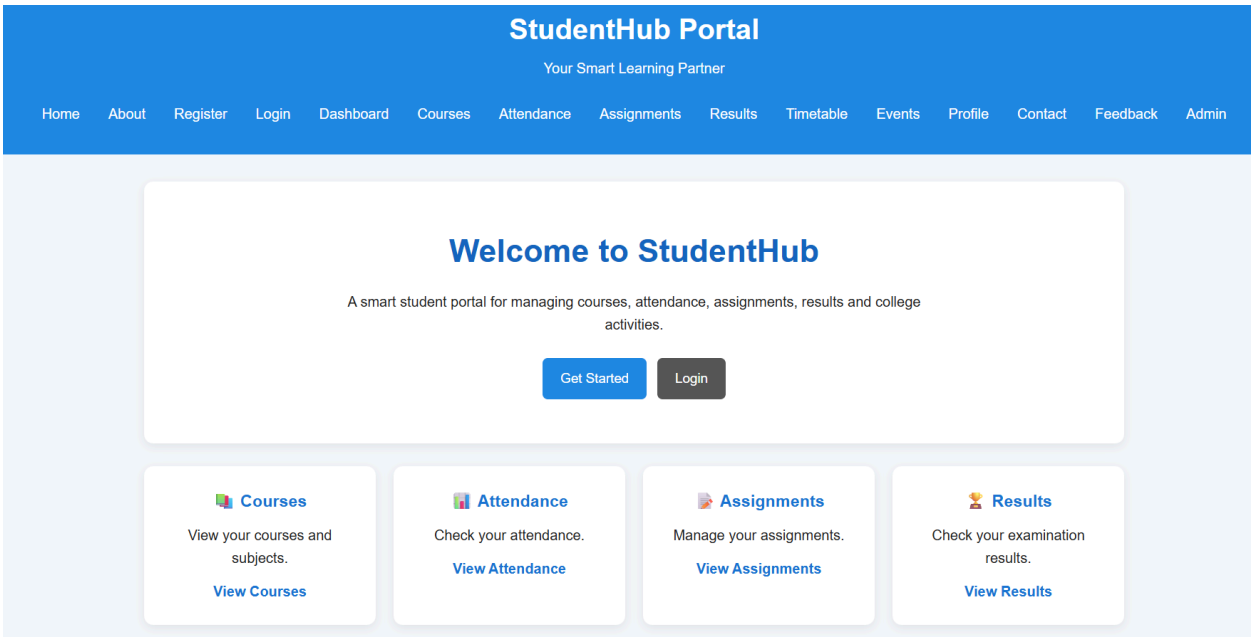
Answer: For searching, filtering, sorting, and pagination, different JavaScript array methods are used.
- **Search:** find() or findIndex()
- **Filter:** filter()
- **Sort:** sort()
- **Pagination:** slice()

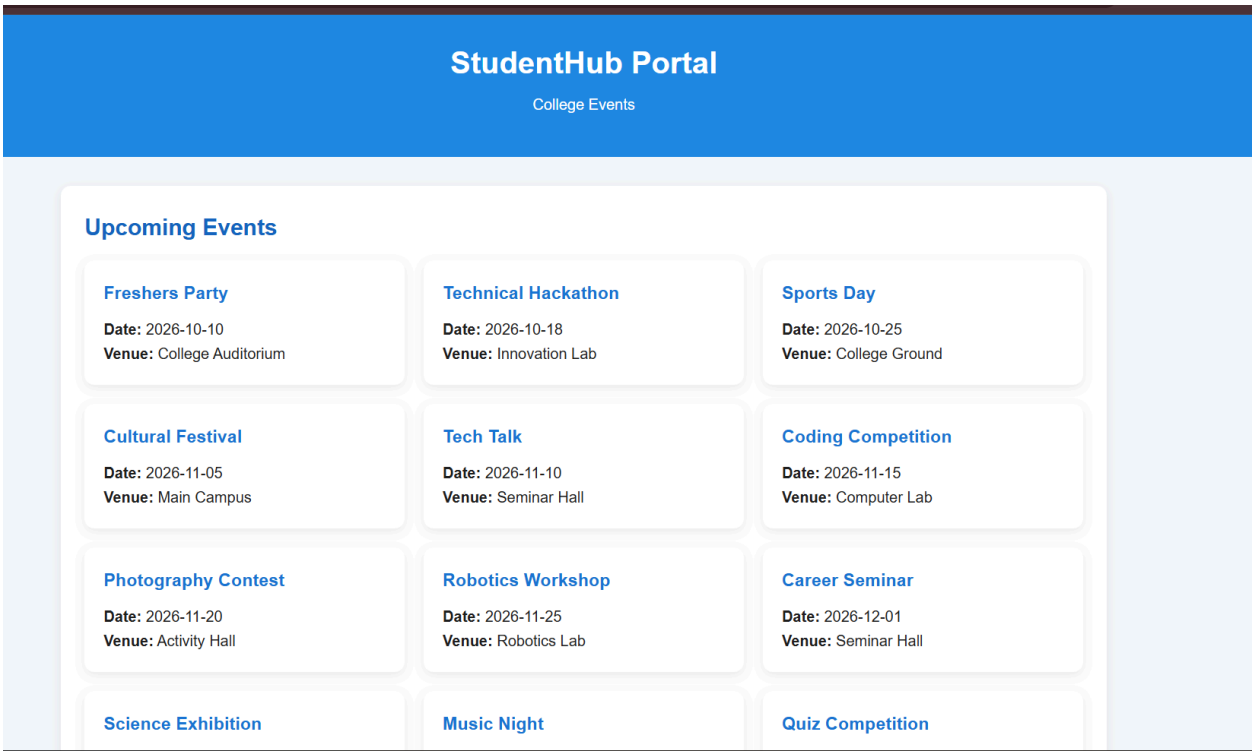
These methods help to easily manage and display JSON data on the webpage.

3. How are loading and error states handled?
Answer: A loading message is shown while fetching data. If fetching is successful, the data is displayed. If an error occurs, an error message is shown using catch().

4. How is modularity maintained in JavaScript files?
Answer: Modularity is maintained by dividing JavaScript code into separate files and functions, where each file handles a specific task.

Result:
Evidence of Work (Screenshots)


StudentHub Portal
Your Smart Learning Partner
Home About Register Login Dashboard Courses Attendance Assignments Results Timetable Events Profile Contact Feedback Admin
Welcome to StudentHub
A smart student portal for managing courses, attendance, assignments, results and college activities.
Get Started
Login
Courses
View your courses and subjects.
View Courses
Attendance
Check your attendance.
View Attendance
Assignments
Manage your assignments.
View Assignments
Results
Check your examination results.
View Results


StudentHub Portal
College Events
Upcoming Events
Freshers Party
Date: 2026-10-10
Venue: College Auditorium
Technical Hackathon
Date: 2026-10-18
Venue: Innovation Lab
Sports Day
Date: 2026-10-25
Venue: College Ground
Cultural Festival
Date: 2026-11-05
Venue: Main Campus
Tech Talk
Date: 2026-11-10
Venue: Seminar Hall
Coding Competition
Date: 2026-11-15
Venue: Computer Lab
Photography Contest
Date: 2026-11-20
Venue: Activity Hall
Robotics Workshop
Date: 2026-11-25
Venue: Robotics Lab
Career Seminar
Date: 2026-12-01
Venue: Seminar Hall
Science Exhibition
Music Night
Quiz Competition

- **GitHub Repository Link:**

https://github.com/25dce015-heet/Web-Development-WDF-

- **Challenges Faced:**

- I faced some difficulty in fetching and reading the JSON data correctly.
- It was a difficult to apply search, filter, and sorting together.like how to update html code and how to apply search operation and fetching.
- I faced problems while implementing pagination for the data.
- Handling loading and error messages was also a challenge.
- Managing the JavaScript code in different files was sometimes confusing.also wrote javascript for this is also difficult .

C.Learning Outcome:-

I learned how to fetch and display JSON data using Fetch API.I also learned search, filter, sorting, and pagination using JavaScript.I understood how to handle loading and error states.

Faculty Remarks:

25DCE015